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Partial Solutions to Universal Problems (APA Version): A Fundamental Introduction

Bachelor Thesis

to obtain the academic degree of

Bachelor of Science in Engineering (BSc)

Bachelor's degree program
Universal Computing
University of Applied Sciences Upper Austria
Hagenberg

Year of submission: 2027

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Affidavit

I hereby declare that I have written the presented thesis independently and have used only the references and aids indicated. All passages taken verbatim or paraphrased from the references are clearly identified as such. This thesis has not been submitted, in whole or in part, in the same or a similar form, to any other examination authority, and is identical to the electronically submitted text document.

Hagenberg, July 1, 2027

Alex A. Wiseguy

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Preface

This document uses the APA citation and reference style (see Ch. 6 for details).

Abstract

This should be a 1-page (maximum) summary of your work in English.

Kurzfassung

An dieser Stelle steht eine Zusammenfassung der Arbeit, Umfang max. 1 Seite. ...

Chapter 1

Introduction

Chapter 2

Writing a Thesis

Chapter 3

Working with LaTeX

Chapter 4

Figures, Tables, Source Code

Chapter 5

Mathematical Elements, Equations and Algorithms

Chapter 6

Using Literature and other Resources

The APA citation style[1] requires different citation macros, depending on the usage and the occurrence of the reference in the text.

6.1 Narrative Citations

For narrative citations, the reference is used as the subject or object of the sentence. The year is placed after the author's name in parentheses. To create this kind of citation use

`\textcite{`*keys*`}`.

Example: Daniel et al. (2018) give a brief introduction to LaTeX, whereas Kopka and Daly (2003) and Oetiker et al. (2021) go into more detail.

6.2 Narrative Citations Within Parentheses

If a reference is to be used *within* parentheses, these must be omitted from the citation. The associated command is

`\nptextcite{`*keys*`}`.

Example: In any case, it is recommended to obtain literature on the topic of LaTeX (e.g., Daniel et al., 2018; Kopka & Daly, 2003; Oetiker et al., 2021).

6.3 Parenthetical Citations

Parenthetical citations are used when the reference should be cited at the end of a sentence or statement. The author's name and year are enclosed in parentheses and separated by a comma. In this case, use

`\parencite{`*keys*`}`.

Example: For LaTeX, there are short introductions (Daniel et al., 2018), as well as more comprehensive works (Kopka & Daly, 2003; Oetiker et al., 2021).

[1] https://apastyle.apa.org/style-grammar-guidelines/references/

Chapter 7

Printing the Manuscript

Chapter 8

Closing Remarks

Appendix A

Technical Details

Appendix B

Supplementary Materials

List of supplementary data submitted to the degree-granting institution for archival storage (in ZIP format).

B.1 PDF Files

Path: /

thesis.pdf . . . . . Master/Bachelor thesis (complete document)

B.2 Media Files

Path: /media

*.ai, *.pdf . . . . . Adobe Illustrator files
*.jpg, *.png . . . . . raster images
*.mp3 . . . . . audio files
*.mp4 . . . . . video files

B.3 Online Sources (PDF Captures)

Path: /online-sources

Reliquienschrein-Wikipedia.pdf ("Reliquienschrein", 2026)

Appendix C

Questionnaire

Appendix D

LaTeX Source Code

Documentation of Generative AI Tool Usage

Responsible use of the AI tools

Statements on responsible use of AI tools	**Confirmation: Yes**
I/we confirm that the outputs generated by AI tools have been verified for accuracy and corrected where required.	☒
I/we acknowledge that the responsibility for the thesis lies with the author(s), and not with the AI tools or any other person.	☒

Detailed activities for which the tools were used

Activity	**Description**	**Tools used (if any)**	**Description of the tool usage** (type of usage, affected sections of the paper/thesis, etc.)
Idea generation & conceptualization	• Generating ideas, objectives and research questions • Identifying and defining relevant concepts and theoretical models • Development of a conceptual framework for the content	—	—
Literature search and analysis	• Search for relevant literature • Reviewing potentially relevant literature • Summaries of relevant literature	Perplexity	Search for literature on ... (Chapter 2)
Methodology	• Search for an appropriate methodology • Development and adaptation of the methodology to the research question(s)	—	—

Activity	**Description**	**Tools used (if any)**	**Description of the tool usage** (type of usage, affected sections of the paper/thesis, etc.)
Coding	• Creation and documentation of code, algorithms, and software • Testing and debugging of existing code, algorithms, software • Understanding and analysis of existing code, algorithms, and software	—	—
Data collection and analysis	• Collection of primary or secondary data • Qualitative data analysis (including summarizing and coding) • Quantitative data analysis (including statistics) • AI-assisted mathematical, computational, or other formal techniques for modeling, simulation, and analytics	—	—
Interpretation and Validation	• Interpretation of results • Derivation of implications for research and practice • Verification of the overall replicability/reproducibility of the results and other research outputs	—	—
Structuring and planning the text	• Structure of the thesis • Structuring of individual sections of the thesis (e.g., bulletpoint outlines per section)	—	—
Generating the text	• Generation of text for topics in various sections of the thesis (including title and abstract)	—	—
Translating text	• Translation of text written by the author(s) • Translation of text written by others	—	—
Reviewing & editing the text	• Critical review, feedback or revision of the content or grammar of the thesis • Proofreading • Rephrasing or paraphrasing text • Shortening / expanding the text	DeepL Write	Proofreading of Chapters 1–5
Citation Management	• Creation of the reference list • Formatting of the references	—	—

Further use of AI tools:

—

Source reference: Gimpel, H., Dilger, P., Lämmermann, L., Urbach, N. (2023). COGNISANCE: Declaration of Generative AI Tool Usage in Higher Education Tests. Research Center for Information Management, May 22, 2023. https://doi.org/10.17605/OSF.IO/VGXF7. Adapted by the University of Applied Sciences Upper Austria. Licensed under CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0/).

References

Literature

Daniel, M., Gundlach, P., Schmidt, W., Knappen, J., Partl, H., & Hyna, I. (2018, April 8). *LaTeX2e-Kurzbeschreibung*. Version 3.0c. https://mirrors.ctan.org/info/lshort/german/l2kurz.pdf (cit. on p. 6).

Kopka, H., & Daly, P. W. (2003). *Guide to LaTeX* (4th ed.). Addison-Wesley. (Cit. on p. 6).

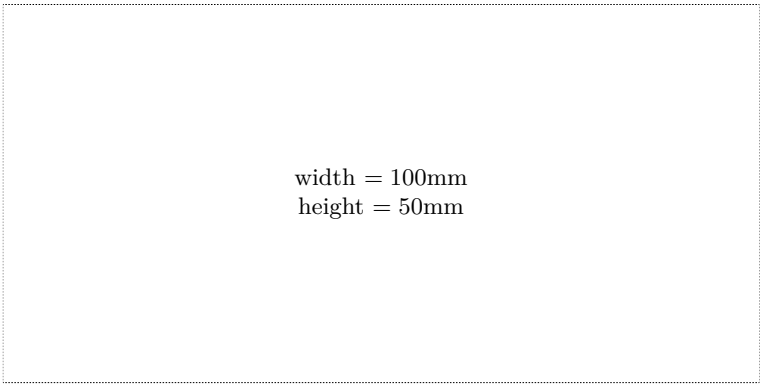
Oetiker, T., Partl, H., Hyna, I., & Schlegl, E. (2021, March 9). *The not so short introduction to LaTeX 2ε: Or LaTeX 2ε in 139 minutes*. Version 6.4. https://mirrors.ctan.org/info/lshort/english/lshort.pdf (cit. on p. 6).

Online sources

Reliquienschrein. (2026, June 10). Retrieved September 13, 2026, from https://de.wikipedia.org/wiki/Reliquienschrein

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